Brendan Boyd

boyd.brendan@stonybrook.edu
biboyd.github.io

Education

SUNY Stony Brook University 2020-Present
Physics & Astronomy Department
PhD Candidate in Physics, with
Concentration in Astronomy

Michigan State University 2016-2020
College of Natural Science, Honors College
Bachelor of Science, Astrophysics
Minors in Math and CMSE

Research & Teaching Experience

Computational Type Ia Supernovae Modeling 2022-Present
Dissertation Research using a low Mach reactive hydrodynamic code to study the convective Urca process in relation to type Ia supernovae.
Multiple publications in the Astrophysical Journal (ApJ)

Teaching Assistant 2020-2021

Computational Galactic Modeling 2019-2020
Developed SALSA Python package for analyzing galaxy simulations via "synthetic observations".
Publication in the Journal for Open Source Software (JOSS)

Undergraduate Learning Assistant 2019-2020

Computational Skills

Programming Skills:
- Proficient in Python
- Proficient in C++
- Competent in FORTRAN
- Competent in HPC workflows (e.g. SLURM, bash scripting)
- Competent in OpenMP threading
- Competent in MPI parallelism

Computational Fluid Modeling
- Low Mach Flows
- (Nuclear) Reactive Flows

Software Projects - Contributor/Developer

MAESTROeX - github.com/AMReX-Astro/MAESTROeX 2022-Present
pynucastro - github.com/pynucastro/pynucastro 2022-Present
initial models - github.com/AMReX-Astro/initial models 2022-Present
SALSA- github.com/biboyd/SALSA 2019-2021